

Impact of Missense Variants Is Hindered by Two Types of Circularity

Dominik G. Grimm, Chloé-Agathe Azencott, Fabian Aicheler, Udo Gieraths, Daniel G. MacArthur, Kaitlin E. Samocha, David N. Cooper, Peter D. Stenson, Mark J. Daly, Jordan W. Smoller, L. E. Duncan, Karsten M. Borgwardt

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Abstract

Comparative evaluations of tools predicting the pathogenicity of missense variants are often confounded by two types of circularity. Type 1 circularity occurs when variants used in training are also included in evaluation benchmarks. Type 2 circularity occurs when tools exploit gene-level properties rather than variant-specific impacts, causing benign variants in disease genes to be falsely labeled as pathogenic. This study provides formal definitions and controlled experimental methodologies to diagnose and prevent circularity in genomic and computational benchmark pipelines.

Role in Our VLM-DB Study

In our research (RQ1 Circularity), Grimm et al.'s circularity taxonomy provides the direct theoretical framework for identifying and eliminating circularity between multimodal feature generation (VLM captioning) and evaluation queries, proving that non-circular evaluation collapses artificial performance gains.